

Notes

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0.0.1 Laboratory Notes

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```
[4]: import numpy as np
import matplotlib.pyplot as plt
```

Task 01 After initially setting up the measurement the unfiltered voltage level on the detector reached $V_{\text{init,RMS}} = 9.11\text{V}$. The noise level of the room was around $\sigma_V = 61.0\text{mV}$. After implementing the polarizers, we turned the first polarizer, since the laser output is somewhat linearly polarized, so we sought the best alignment of the first filter with it. Firstly our angle readout error is significant, the thickness of the pointer is around two degrees wide. Moreover, the background lighting fluctuations due to our own shadows yielded a net $\pm 20\text{mV}$ error on all of our measurements.

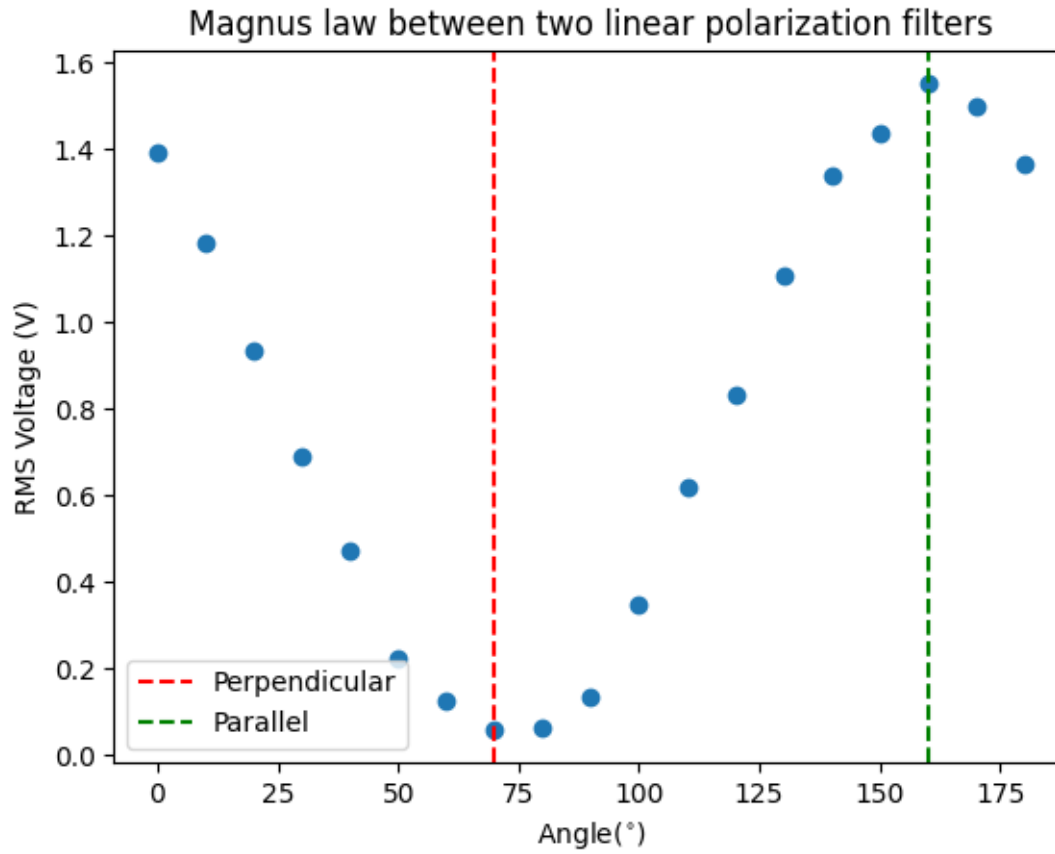
```
[12]: angles = np.linspace(0, 180, 19) # in degrees
print(angles)
voltage= np.array([1.392, 1.185, 0.934, 0.690, 0.470, 0.221, 0.126, 0.058, .
    ↪064, .135, .347, .618, .830, 1.107, 1.339, 1.435, 1.553, 1.498, 1.364]) # in
    ↪V
plt.figure()
plt.scatter(angles, voltage)
plt.title('Magnus law between two linear polarization filters')
plt.xlabel(r'$\mathrm{Angle}(\text{\textcircled{}})$')
plt.ylabel('RMS Voltage (V)')

min_angle = angles[np.argmin(voltage)]
max_angle = angles[np.argmax(voltage)]

plt.axvline(x=min_angle, color='red', linestyle='--', label='Perpendicular')
plt.axvline(x=max_angle, color='green', linestyle='--', label='Parallel')
plt.legend()

plt.show()
```

```
[ 0.  10.  20.  30.  40.  50.  60.  70.  80.  90. 100. 110. 120. 130.
 140. 150. 160. 170. 180.]
```



Task 02 Starting from the second task we have closed the shaders on the window. The new noise level is $\sigma_V = 50\text{mV}$. As the previous measurement suggests the parallel orientation of the two filters was at an angle of 160° . With parallel polarizers the maximal intensity was detected at exactly (with readoff precision, of 3°) 0° of LC rotation. After setting the polarizers perpendicular the new maximal LC orientation was indeed at 45° . Measuring the detector voltage we realised a constant voltage level irrespective of the analyzer orientation (within margin of error), which can only mean that the LC circularly polarised the linear beam.

```
[13]: LCangle = np.linspace(0, 180, 19)
LCvoltage = np.array([.451, .471, .501, .510, .505, .512, .499, .480, .461, .
↪431, .425, .425, .415, .415, .423, .420, .458, .480, .496])

theta = np.deg2rad(LCangle)

plt.figure(figsize=(7, 5))
ax = plt.subplot(111, projection='polar')

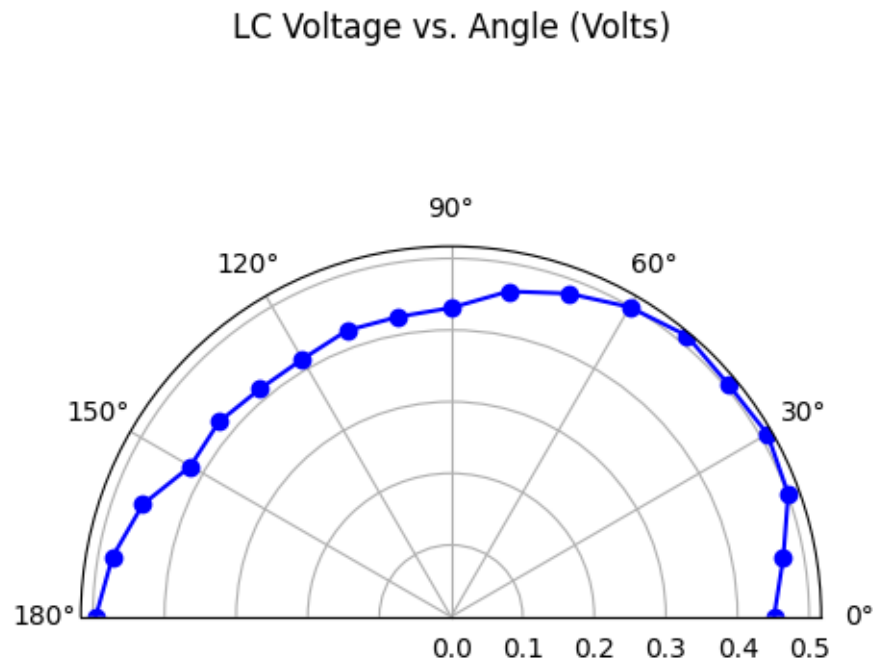
ax.plot(theta, LCvoltage, marker='o', linestyle='-', color='b', label='LC
↪Voltage')
```

```

ax.set_thetamin(0)
ax.set_thetamax(180)
ax.set_title("LC Voltage vs. Angle (Volts)", va='bottom')

plt.show()

```



Task 03

```

[18]: inputvoltage = np.array([0.000, 0.010, 0.050, 0.100, 0.200, 0.300, 0.400, 0.
    ↪500, 0.700, 1.000, 1.100, 1.200, 1.300, 1.400, 1.500, 1.600, 1.700, 1.800, 1.
    ↪900, 2.000, 2.100, 2.200, 2.300, 2.400, 2.500, 2.600, 2.700, 2.800, 2.900, 3.
    ↪000, 3.100, 3.200, 3.300, 3.400, 3.500, 3.600, 3.700, 3.800, 3.900, 4.000, 4.
    ↪100, 4.200, 4.300, 4.400, 4.500, 4.600, 4.700, 4.800, 4.900, 5.000, 5.300, 5.
    ↪600, 5.900, 6.200, 6.500, 6.800, 7.100, 7.400, 7.700, 8.000, 8.500, 9.000, 9.
    ↪500, 10.000])

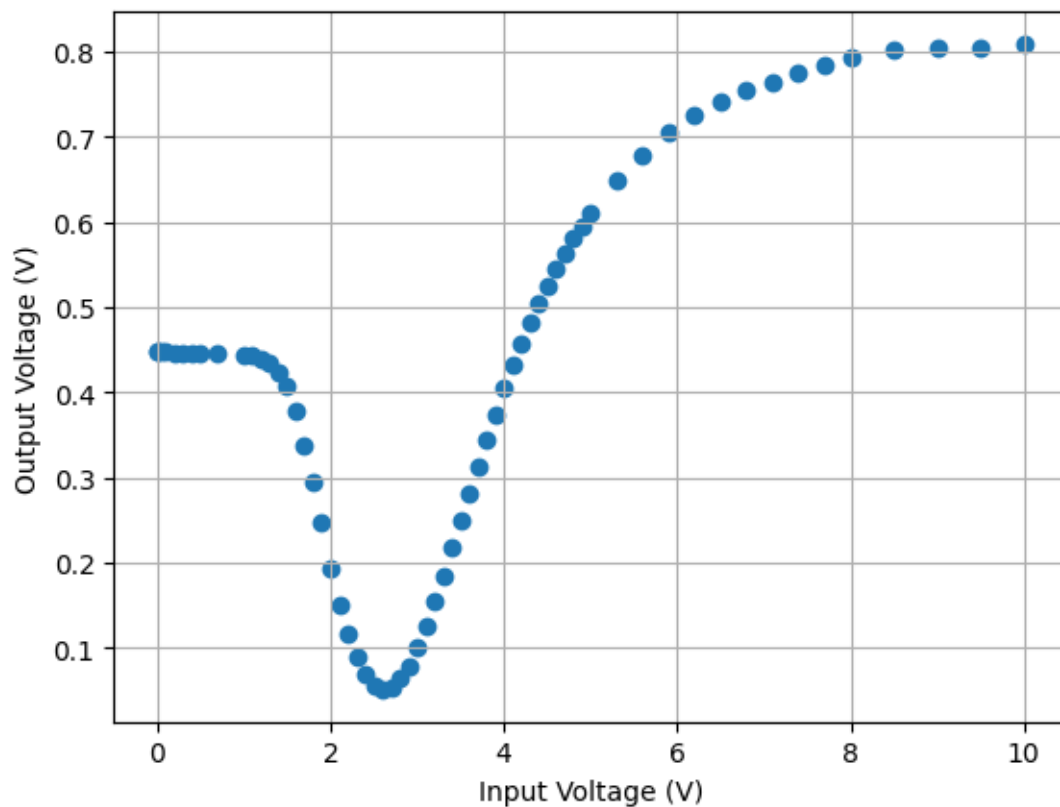
```

```

outputvoltage = np.array([.447, .447, .447, .447, .446, .446, .446, .446, .446,
↪.444, .444, .439, .434, .424, .407, .378, .338, .294, .247, .193, .151, .
↪.117, .089, .068, .056, .051, .054, .064, .079, .100, .125, .154, .184, .217,
↪.249, .281, .313, .343, .374, .404, .431, .457, .482, .504, .525, .544, .
↪.563, .580, .595, .611, .648, .679, .704, .725, .741, .754, .763, .775, .785,
↪.792, .801, .805, .804, .809])

plt.figure()
plt.scatter(inputvoltage, outputvoltage)
plt.xlabel('Input Voltage (V)')
plt.ylabel('Output Voltage (V)')
plt.grid(True)
plt.show()

```



Task 04

[22]:

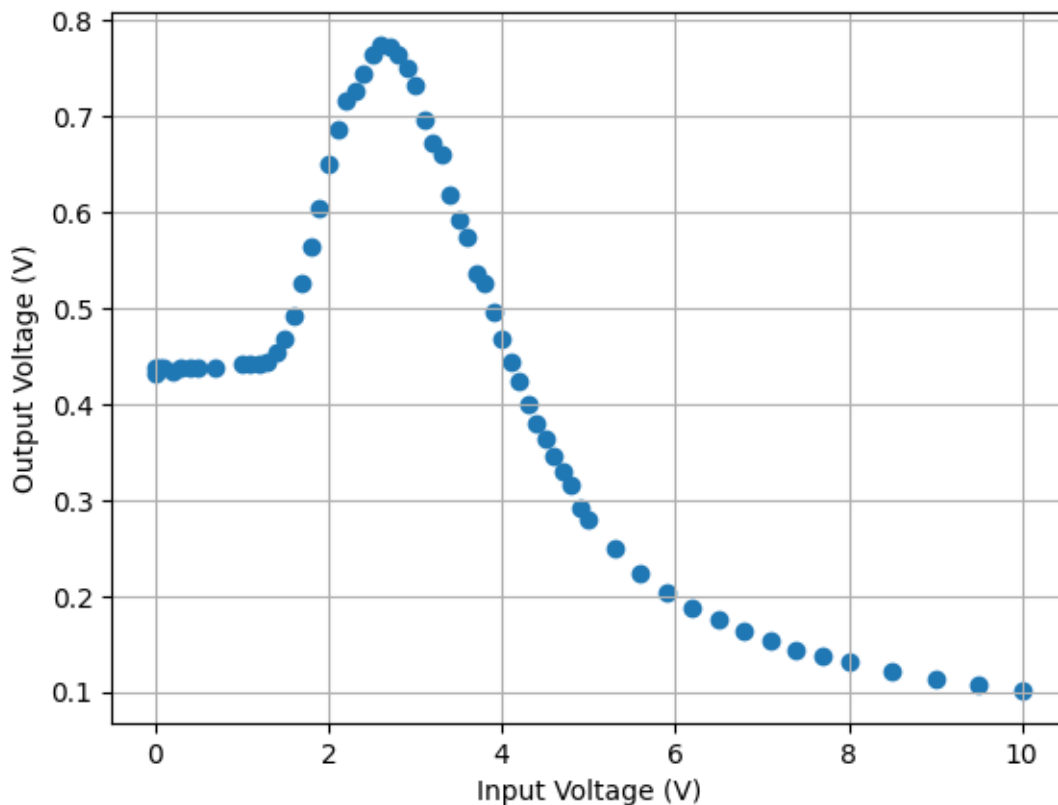
```

InputVoltage=np.array([0.000, 0.010, 0.050, 0.100, 0.200, 0.300, 0.400, 0.500,
↪0.700, 1.000, 1.100, 1.200, 1.300, 1.400, 1.500, 1.600, 1.700, 1.800, 1.900,
↪2.000, 2.100, 2.200, 2.300, 2.400, 2.500, 2.600, 2.700, 2.800, 2.900, 3.000,
↪3.100, 3.200, 3.300, 3.400, 3.500, 3.600, 3.700, 3.800, 3.900, 4.000, 4.100,
↪4.200, 4.300, 4.400, 4.500, 4.600, 4.700, 4.800, 4.900, 5.000, 5.300, 5.600,
↪5.900, 6.200, 6.500, 6.800, 7.100, 7.400, 7.700, 8.000, 8.500, 9.000, 9.500,
↪10.000])
OutputVoltage=np.array([.433,.439,.439,.439,.434,.438,.438,.438,.438,.442,.443,.
↪442,.445,.454,.469,.493,.527,.564,.604,.650,.686,.717,.727,.745,.765,.775,.
↪773,.765,.751,.733,.696,.672,.660,.619,.592,.575,.537,.526,.496,.469,.444,.
↪424,.400,.381,.365,.347,.331,.316,.293,.280,.250,.225,.203,.188,.175,.163,.
↪153,.144,.137,.131,.122,.113,.107,.102])
print(len(InputVoltage))
print(len(OutputVoltage))
plt.figure()
plt.scatter(InputVoltage, OutputVoltage)
plt.xlabel('Input Voltage (V)')
plt.ylabel('Output Voltage (V)')
plt.grid(True)
plt.show()

```

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Concerning the shapes of the plots the expected results showed up, since the second one is the inverted copy of the first, which is reasonable since it should be the same curve subtracted from π showing an inverted shape. Saturation is observed in both cases at high voltages.